MAAZ BIN IRFAN

Computer Science Undergraduate • Software Engineer • Automation & Homelab Engineer

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PROFESSIONAL SUMMARY

Results-driven BS Computer Science student (4th Semester) combining core programming in **Python, C++, and C#** with high-velocity full-stack engineering, complex **workflow automation**, and hands-on Linux administration. Proven track record of architecting and shipping production-ready systems—including the **Raasta** real-time ride-hailing mobile platform, **TrendRadar PK** automated 35-node n8n e-commerce scraper & ads radar, **ApexOutreach** autonomous multi-agent WhatsApp bot, and **DukaanPOS** retail software with 11 published releases—backed by a self-hosted Docker homelab.

TECHNICAL SKILLS

Programming Languages	Python, C++, C#, TypeScript, JavaScript, SQL (PostgreSQL, SQLite), Bash / Shell
Automation & Data Pipelines	n8n Workflow Automation, Playwright, REST APIs, Webhooks, Shopify & Daraz Scrapers, Meta Ads Intelligence
Frontend & Mobile	React Native, Expo (SDK 58), React.js, Vite, HTML5/CSS3 (Glassmorphism UI), PySide6 (Qt)
Backend, DB & Mapping	Node.js, Express.js, .NET Core / WPF, Supabase (PostgreSQL, Realtime, RLS), MapLibre GL, Project-OSRM
DevOps, Linux & Homelab	Linux Administration (Debian, Ubuntu, Fedora), Docker & Docker Compose, Self-Hosting, Git/GitHub, systemd
Core CS & Methodologies	Data Structures & Algorithms (DSA), OOP, Database Systems, Concurrency, Agentic Dev Workflows

FEATURED TECHNICAL & AUTOMATION PROJECTS

Raasta — P2P Ride-Hailing, Parcel Delivery & Travel Mobile App

Lead Developer

React Native, Expo (SDK 58), TypeScript, Supabase (PostgreSQL, Realtime), MapLibre GL, Project-OSRM

- Architected an inDrive-style ride-hailing and parcel delivery cross-platform mobile app featuring real-time passenger-driver fare bidding and negotiation.
- Engineered a **100% free, zero-billing mapping stack** using MapLibre GL, OpenFreeMap vector tiles, Nominatim geocoding, and Project-OSRM road routing polylines and ETA calculation, saving thousands in Google Maps/Mapbox API billing.
- Implemented sub-second driver location tracking, ride request broadcasting, and driver-passenger chat via Supabase Realtime with Row-Level Security.

TrendRadar PK — Pakistan E-Commerce & Meta Ads Automation Radar

n8n & Automation

Self-Hosted n8n (35 Nodes), Supabase (PostgreSQL), Vite, JavaScript, Apify, Shopify & Meta APIs

- Designed and deployed an automated **35-node n8n workflow pipeline** executing daily scheduled scraping of Pakistani Shopify stores and Daraz.pk product catalogs, pricing, and velocity.
- Engineered a Meta Facebook Ads scraper for Pakistan (`country=PK`) to track winning ad creatives, monitoring **Ad Longevity**, **Uptime Streak**, and **Takedown/Restart Frequency**.
- Built custom momentum scoring inside n8n Code nodes, syncing deduplicated winning items to Supabase and rendering them on a responsive dark glassmorphism dashboard.

ApexOutreach — Autonomous Multi-Agent Lead Gen & WhatsApp Outreach Bot

AI Agents & Python

Python, Playwright, PySide6 (Qt), SQLite, Asyncio, Bayesian Multi-Armed Bandit, PyInstaller

- Built an autonomous multi-agent pipeline with specialized agents for Google Maps lead scraping, FMCSA SaferWeb US carrier verification (USDOT/MC), and E.164 phone sanitization.
- Automated direct WhatsApp Web delivery using Playwright with human-like typing simulation, QR authentication, and anti-ban jitter delays.
- Integrated a self-learning **Bayesian Multi-Armed Bandit** reinforcement loop to dynamically score and optimize message templates based on positive response rates; shipped as a standalone PySide6 Windows executable.

DukaanPOS — Enterprise Desktop Retail POS & Inventory Management

Desktop & Cloud

C#, .NET, WPF, SQLite, Entity Framework Core, ZXing.Net, Firebase Sync, InnoSetup

- Engineered an enterprise POS system deployed in active retail environments, featuring rapid barcode checkout, held transactions, inventory tracking, and ESC/POS thermal receipt printing using vector barcodes (ZXing).
- Implemented an offline-first SQLite data architecture with EF Core migrations, concurrency resolution, and Firebase cloud backup; released 11 production versions (v1.0.1–v1.0.11) via custom InnoSetup installers.

Self-Hosted Docker Homelab & Linux Server Infrastructure

DevOps & Homelab

Linux (Debian / Fedora / Ubuntu), Docker & Docker Compose, systemd, Bash, UFW, Self-Hosting

- Provision and maintain a bare-metal Linux homelab server running Docker Compose to host isolated services, including production self-hosted n8n instances and PostgreSQL databases.
- Configured isolated bridge networks, persistent volume backup routines, systemd daemons, firewall rules, and automated bash maintenance cron scripts.

EDUCATION

Bachelor of Science in Computer Science (BSCS) — 4th Semester

2024 – 2028 (Expected)

Govt. Khawaja Fareed Graduate (Postgraduate) College, Rahim Yar Khan • Affiliated with The Islamia University of Bahawalpur (IUB)

Relevant Coursework: Programming Fundamentals, Object-Oriented Programming (C++/Java), Data Structures & Algorithms, Database Systems, Operating Systems, Digital Logic Design.